

→ First,

* - Dependent Events

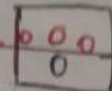
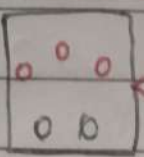
A = Event₁

B = Event₂

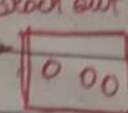
say, Marbles {Red, Black} → 5 Total.

(2/5)

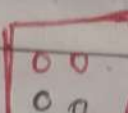
= P(A) Block was picked



P(B/A) → P(Block out of Red Marbles)



2 Black



$$P(A \cap B) = \frac{2}{5} \times \frac{1}{4} \Rightarrow \frac{2}{20} = \frac{1}{10}$$

$$\frac{2}{20} = \frac{1}{10}$$

* Independent Events

$$\rightarrow P(A)$$

say, Black Balls = 1 Red, 1 Green

$$P(A_{\text{Black}}) = \frac{1}{3} \Rightarrow 0.33$$

Conditional Probability

$$* P(A/B) = \frac{P(A \cap B)}{P(B)}$$

say, for above Ex.

$$\left[P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \cap B)}{2/5} \right]$$

$$P(A/B) = P(A \cap B) / P(B)$$

$$P(A/B) * P(B) = P(A \cap B)$$

$$\Rightarrow \frac{1}{4} \times \frac{2}{5} \Rightarrow \frac{1}{10} = \frac{1}{10} \text{ } \checkmark$$

say,

$$P(A/B)$$

↳ Probability of A given B has already occurred.

$$P(A \cap B) = P(A/B) * P(B)$$

$$P(B \cap A) = P(B/A) * P(A)$$

$$P(A/B) * P(B) = P(B/A) * P(A)$$

$$P(A/B) = \frac{P(A)}{P(B)} * P(B/A)$$

Diagram illustrating the components of the equation:

- $P(A/B)$ is labeled as Posterior Probability.
- $\frac{P(A)}{P(B)}$ is labeled as Prior.
- $P(B/A)$ is labeled as Likelihood.
- $P(A)$ is labeled as Marginal.